Investigation of Hot Carrier Injection During GIDL Erase Operation in 3-D NAND Flash Memory

Jae Hyun Nam, Jin Ho Chang, *Graduate Student Member, IEEE*, Kyung Moon Kim, Jin Wook Lee, *Graduate Student Member, IEEE*, Da Eun Yang, Suk-Kang Sung, Jun Hyoung Kim, and Woo Young Choi, *Senior Member, IEEE*

***Abstract*— The physical mechanism and transient behavior of hot carrier injection (HCI) during the gate-induced drain leakage (GIDL) erase operation in 3-D NAND flash memory devices are systematically investigated based on both experimental and simulation data. The transient analysis of HCI focuses on the evolution of two key physical parameters: the potential difference between the source line/bitline (SL/BL) and the string, and the maximum lateral electric field in the select gate transistor (SG) channel. It is found that a slow rise of the string potential, resulting from the accumulation of GIDL-generated holes in the floating string, induces significant HCI at SG. The dependence of HCI on critical design parameters, including the number of wordline (WL) layers, SL/BL voltage rise time, offset length, SG gate bias, and its rise time, is also analyzed, revealing a strong correlation between HCI behavior and GIDL efficiency. The presented systematic analysis provides physical insight and design guidelines for mitigating HCI-induced SG degradation, thereby enhancing the reliability of state-of-the-art 3-D NAND flash memory technologies.**

***Index Terms*— 3-D NAND flash memory, gate-induced drain leakage (GIDL) erase operation, hot carrier injection (HCI).**

I. Introduction

THE prominence of 3-D NAND flash memory, which has become a mainstream storage solution owing to its high bit density, has continued to grow with increasing demands for data-intensive applications such as artificial intelligence (AI) training and cloud computing [1]. To respond to this trend, the cell-over-peripheral (COP) structure has been introduced, in which memory arrays are placed over the peripheral circuits. It enhances bit density while remaining relatively immune to reliability and performance degradation compared with other downscaling methods, making it widely adopted in modern 3-D NAND technologies [2], [3].

However, the COP structure necessitates a shift in the erase operation scheme of 3-D NAND. The conventional body erase scheme is not applicable, as the string is connected to an n^+ polysilicon plate instead of a p-type silicon substrate. As a result, holes must be generated in select gate transistors (SGs) located at both ends of the string through the gate-induced drain leakage (GIDL) mechanism. The generated holes accumulate in the floating string, raising the string potential (V_B) required for cell erase in a process referred to as the "GIDL erase scheme" [4], [5].

In the GIDL erase scheme, high source line (SL) and bitline (BL) voltages ($V_{SL/BL}$) are applied to the SGs to generate sufficient holes for reliable erase operation [6], which in turn induces unintended hot carrier injection (HCI) because V_B cannot rise as rapidly as $V_{SL/BL}$. The resulting difference between $V_{SL/BL}$ and V_B induces a strong lateral electric field in the SG channel, subsequently triggering HCI. This HCI causes a threshold voltage (V_{th}) shift in the SGs, degrading the 3-D NAND functionality by increasing power consumption, slowing operation speed, and raising program/read disturbances because the SGs play an essential role in string-selective operation within the memory array [7], [8]. Thus, managing the V_{th} window of the SGs under the HCI stress during GIDL erase is a crucial reliability issue.

Although extensive studies have been conducted on HCI in 3-D NAND [9], [10], [11], [12], [13], [14], most studies have either overlooked the impact of HCI during erase operations or focused only on the conventional body erase schemes. The transient behavior of erase-induced HCI under GIDL erase conditions remains largely unexplored. Moreover, because SGs are directly involved in the hole generation process during the GIDL erase operation, a strong correlation exists between HCI

Received 4 November 2025; revised 22 December 2025; accepted 31 December 2025. Date of publication 14 January 2026; date of current version 26 February 2026. This work was supported in part by Samsung Electronics, in part by the Ministry of Science and ICT (MSIT) through the Development of Flash Memory-Based Artificial Intelligence (AI) Processing Unit for On-Device AI under Grant RS-2025-02314443, in part by the Mid-Career Researcher Program under Grant NRF-2021R1A2C1007931, in part by the Artificial Intelligence Semiconductor Support Program under Grant IITP-2023-RS-2023-00256081, and in part by the Korea Institute for Advancement of Technology (KIAT) funded by the Korea Government (MOTIE) under Grant RS-2025-02214408. The review of this article was arranged by Editor S. Raghunathan. *(Corresponding author: Woo Young Choi.)*

Jae Hyun Nam, Jin Ho Chang, Kyung Moon Kim, Jin Wook Lee, Da Eun Yang, and Woo Young Choi are with the Department of Electrical and Computer Engineering and the Inter-University Semiconductor Research Center (ISRC), Seoul National University, Gwanak-gu, Seoul 08826, Republic of Korea (e-mail: wooyoung@snu.ac.kr).

Suk-Kang Sung and Jun Hyoung Kim are with the Advanced Flash Technology Team, Samsung Electronics Company Ltd., Hwaseong 18448, Republic of Korea.

Digital Object Identifier 10.1109/TED.2026.3651215

1557-9646 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Korea University. Downloaded on June 03,2026 at 07:47:29 UTC from IEEE Xplore. Restrictions apply.

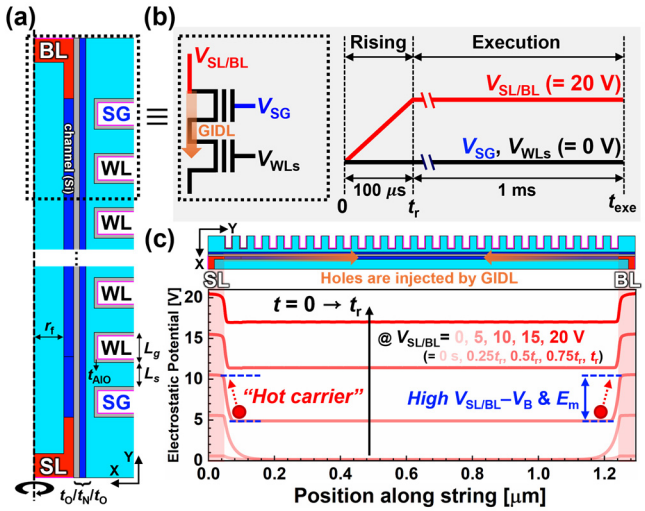


Fig. 1. (a) Cylindrical cross-sectional view of a 3-D NAND flash memory string. (b) Schematic of the select gate transistor (SG) and the bias scheme for the GIDL erase operation. (c) Evolution of the string potential (V_B) during the rising phase.

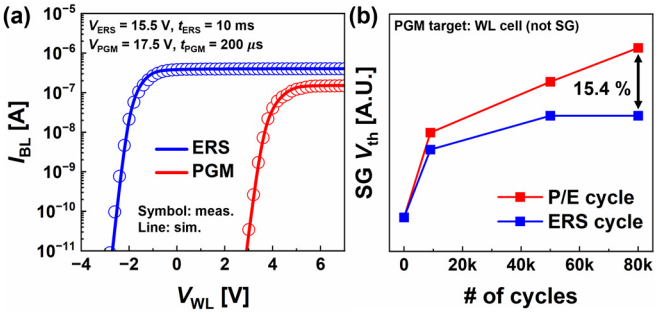


Fig. 2. Experimental 3-D NAND flash memory data. (a) I_{BL}–V_{WL} curves of a memory cell: the median value of the different cells after a one-shot program/erase operation. (b) V_{th} shifts of the SG as a function of program/erase and erase-only cycling number.

and erase efficiency. Therefore, a detailed analysis is required to develop effective HCI mitigation strategies compared to the conventional body erase scheme.

In addition, recent studies have proposed erase-intensive operation schemes for 3-D NAND, such as subblock-level erase and incremental step pulse erase (ISPE) [15], [16], [17]. These schemes increase both the number of erase operations and the associated cycle stress, further highlighting the importance of understanding erase-induced HCI and its impact on reliability.

This study provides a systematic analysis of HCI during the GIDL erase operation in 3-D NAND for the first time. The physical mechanism and transient behavior under the floating string condition are analyzed in terms of the evolution of two key physical parameters. Furthermore, the dependence of HCI on various design parameters is examined, providing physical insights and design guidelines for HCI mitigation.

II. HCI Phenomenon During GIDL Erase

Fig. 1(a) shows the cross section of a simulated 3-D NAND device with a cylindrical macaroni structure, which includes two SGs, wordlines (WLs), an SL, and a BL. Fig. 1(b) shows the schematic of the SG and the bias scheme for the GIDL erase operation. A high $V_{SL/BL}$ is applied to the SGs to generate holes through the GIDL mechanism. To raise V_B, both sufficient hole generation in the SGs and the charging of the string capacitance (C_B) are required [18]. However, due to the low initial hole generation rate and the large C_B, V_B cannot rise as rapidly as $V_{SL/BL}$, resulting in a significantly high potential difference ($V_{SL/BL} - V_B$) during the rising phase, as shown in Fig. 1(c). This high $V_{SL/BL} - V_B$ induces a strong lateral electric field in the SG channel, which subsequently triggers an unintended HCI. In the typical dual-sided injection scheme, where GIDL occurs at both the SL and BL sides uniformly, V_B remains nearly constant along the string owing to the short hole propagation time, except in the depletion regions located at both ends. These depletion regions are confined near the SL/BL junctions during the erase operation as a result of the continuous accumulation of generated holes in the string, causing the HCI to be primarily localized adjacent to the SL/BL junctions.

TABLE I
Summary of Simulated Device Parameters

Cell gate length (L_g)	25 nm	Filler oxide radius (r_f)	25 nm
Cell space length (L_s)	20 nm	Channel doping concentration (N_A^B)	10^{14} cm^{-3}
Alumina thickness (t_{AlO})	3 nm	SL/BL doping concentration ($N_D^{SL} = N_D^{BL}$)	10^{20} cm^{-3}
O/N/O thickness ($t_{O/N/O}$)	5/5/7 nm	Gate metal work function (φ_m)	4.5 eV
Channel thickness (t_{ch})	8 nm	Number of WLs (N_{WL})	27

To investigate this HCI phenomenon during the GIDL erase operation, we used a commercial device simulator [19], calibrating the TCAD deck using measurement data of the conventional 3-D NAND flash memory [20]. Fig. 2(a) shows the simulated I_{BL}–V_{WL} characteristics of the WL cell, which are in good agreement with the experimental data. Fig. 2(b) shows that V_{th} shifts of the SG experiencing the program/erase and erase-only stress conditions. As previously discussed, GIDL erase-induced HCI leads to the significant V_{th} shift of SG. The simulation solves the Poisson equation in cylindrical coordinates and incorporates a nonlocal band-to-band tunneling (BTBT) model, the lucky electron model, and the van Overstraeten–de Man impact ionization model. The device parameters of the simulated 3-D NAND are summarized in Table I.

Fig. 3(a) shows the spatial distribution of HCI-induced trapped electrons in the charge trap layer (CTL). These trapped electrons are primarily generated as secondary hot electrons through impact ionization. Owing to the macaroni structure with a virtual junction in 3-D NAND, the longitudinal BTBT current along the string direction becomes the dominant component during the erase operation, making the SG behave similar to a reverse-bias gated p–n diode [21]. Under this condition, the high lateral electric field near the SL and

Authorized licensed use limited to: Korea University. Downloaded on June 03,2026 at 07:47:29 UTC from IEEE Xplore. Restrictions apply.

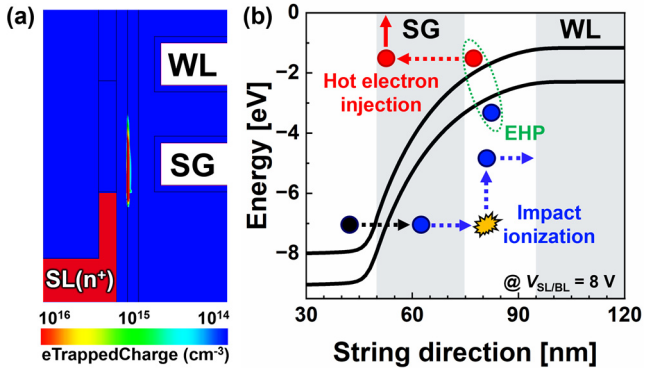


Fig. 3. (a) Spatial distribution of HCI-induced trapped electrons in the CTL. (b) Band diagram in the longitudinal direction of the string. A strong lateral electric field induces BTBT, and the generated carriers trigger impact ionization, where some of the secondary hot electrons are injected into the CTL of SGs.

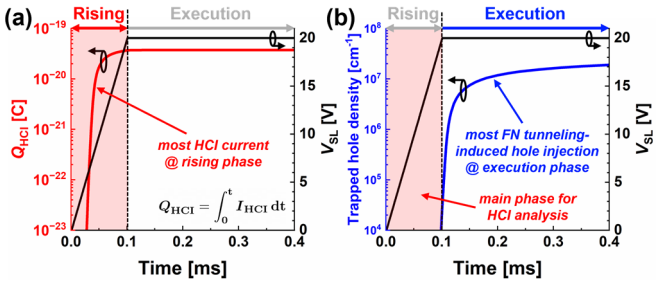


Fig. 4. Transient behavior of (a) total injected electron charge (Q_{HCI}) and (b) FN tunneling-induced trapped hole density in the SG CTL with the V_{SL} pulse during the erase operation.

BL junctions induces BTBT, and the generated holes trigger impact ionization, leading to the injection of secondary hot electrons, as shown in Fig. 3(b). This HCI current (I_{HCI}) is governed by $V_{SL/BL} - V_B$ and the maximum lateral electric field in the SG channel (E_m) [22], [23], [24]. Therefore, the subsequent transient analysis of HCI focuses on the evolution of these two parameters. Due to the symmetrical characteristics of the SL and BL junctions, the subsequent analysis focuses solely on the region adjacent to the SL junction.

III. Analysis of Transient HCI Behavior

During the erase operation, particularly during the rising phase, the GIDL-induced hole generation rate and the resulting string potential change over time. These transient conditions lead to a distinct evolution of $V_{SL} - V_B$ and E_m, necessitating a comprehensive understanding of the transient behavior of HCI to gain physical insight and establish effective mitigation strategies. Fig. 4(a) shows the total HCI-induced injected electron charge (Q_{HCI}) during the erase operation. Most of Q_{HCI} is generated during the initial rise of the V_{SL} pulse and then saturates over the remainder of the pulse. In contrast, as shown in Fig. 4(b), the Fowler–Nordheim (FN) tunneling-induced hole injection into the SG CTL mainly occurs during the execution phase and has a negligible impact on the HCI behavior during the rising phase, which is the primary focus of this study. Moreover, in practical erase operations, FN tunneling

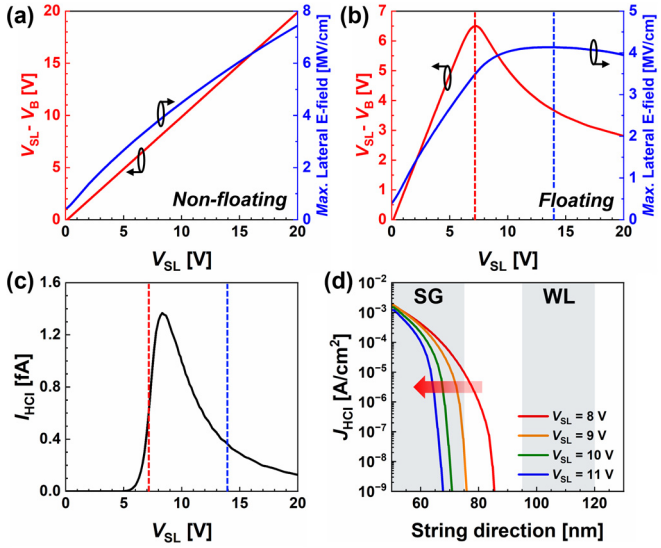


Fig. 5. Evolution of the potential difference ($V_{SL} - V_B$) and the maximum lateral electric field (E_m) under (a) nonfloating condition (V_{SL} = 20 V and V_{BL} = 0 V) and (b) floating condition (V_{SL} = 20 V and V_{BL} = 20 V). Evolution of (c) I_{HCI} and (d) J_{HCI} in the string under the floating condition during the rising phase.

in SG is effectively suppressed by applying the SG gate bias to prevent unintended SG erase. Therefore, the subsequent transient analysis of HCI focuses on the rising phase, with the influence of FN tunneling decoupled in the initial stage to establish the baseline HCI physics and its dependence on design parameters. A comprehensive analysis of the SG V_{th} shift that incorporates both HCI and FN tunneling is presented in Section IV-C.

The distinct transient behavior of HCI originates from the floating string state, where both SGs at the ends of the string remain in the OFF-state due to high $V_{SL/BL}$ bias applied during the erase operation. In addition, the oxide filler of the macaroni structure electrically isolates the string. Fig. 5(a) and (b) compares the evolution of $V_{SL} - V_B$ and E_m during the rising phase under the nonfloating string (general operation) and floating string (GIDL erase operation) conditions. In the nonfloating condition, both parameters increase monotonically with V_{SL}. In contrast, under the floating condition, both parameters initially increase but subsequently decrease after reaching their respective peaks. This result arises from two competing effects. As the V_{SL} pulse rises, the drain-to-gate potential (V_{DG}) of SG increases, enhancing both $V_{SL} - V_B$ and E_m, while simultaneously, holes generated through the GIDL mechanism accumulate in the floating string, raising V_B and thereby reducing both $V_{SL} - V_B$ and E_m.

Based on the transient characteristics of the floating string, the evolution of I_{HCI} during the rising phase is analyzed with respect to the peak points of two key physical parameters, $V_{SL} - V_B$ and E_m, as shown in Fig. 5(c). At the initial stage of the rising phase (ahead of the $V_{SL} - V_B$ peak), the GIDL-induced hole generation rate remains low, resulting in insufficient hole accumulation in the string. Accordingly, both $V_{SL} - V_B$ and E_m increase in a manner similar to the nonfloating string condition, leading to an increase in I_{HCI}. However, I_{HCI}

Authorized licensed use limited to: Korea University. Downloaded on June 03,2026 at 07:47:29 UTC from IEEE Xplore. Restrictions apply.

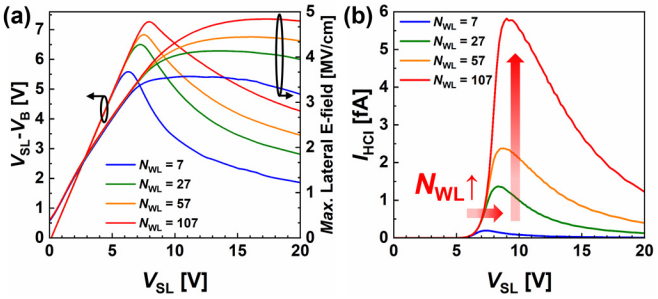


Fig. 6. Evolution of (a) $V_{SL} - V_B$ and E_m and (b) I_{HCI} during the rising phase for different numbers of WL layers (N_{WL}).

remains relatively small due to the limited carrier supply. As the V_{SL} pulse rises, $V_{SL} - V_B$ begins to decrease due to the enhanced hole generation rate, which rapidly raises V_B. Meanwhile, E_m continues to increase because the rising V_B has a weaker influence on E_m than that of V_{SL}. This result is attributed to the wide depletion region between the SL junction and the hole accumulation region in the string, as shown in Fig. 1(c). Although these two opposing effects coexist (decrease in $V_{SL} - V_B$ and increase in E_m), I_{HCI} eventually decreases after reaching its peak. This behavior arises because the accumulated holes in the string continuously reduce the depletion width and narrow the region where impact ionization and HCI occur, as shown in Fig. 5(d). As the V_{SL} pulse rises further (after the peak of E_m), additional hole accumulation continuously narrows the depletion region and increases the influence of V_B on E_m. This results in a decrease in E_m along with $V_{SL} - V_B$, and a continued reduction in I_{HCI}. Consequently, the peak of I_{HCI} occurs between the peaks of $V_{SL} - V_B$ and E_m.

To validate this approach, the same analysis was performed under conditions with an increased number of WL layers (N_{WL}), reflecting the 3-D NAND scaling trend. As shown in Fig. 6(a), increasing N_{WL} shifts the peaks of $V_{SL} - V_B$ and E_m to the right and increases their magnitudes. This behavior originates from increased C_B proportional to N_{WL} [18], requiring greater hole accumulation in the string to raise V_B, as governed by $Q = C_B V_B$. A higher GIDL-induced hole generation rate is needed during the rising phase, leading to a larger $V_{SL} - V_B$ and E_m values. Consequently, I_{HCI} increases, and the corresponding peak V_{SL} value shifts to a higher range, as shown in Fig. 6(b). These results confirm that the proposed analysis is consistently applicable under scaling conditions and further highlight that the GIDL erase-induced HCI becomes an increasingly critical reliability concern as 3-D NAND continues to scale with larger N_{WL}.

IV. Impact of Design Parameters on HCI

The impact of several key design parameters of 3-D NAND that directly influence HCI is examined to provide physical insight and design guidelines for mitigating HCI. Owing to the direct involvement of SGs in hole generation during the erase process, the GIDL erase scheme establishes a strong correlation between HCI mitigation and GIDL efficiency. This observation necessitates a more detailed analysis to develop effective mitigation strategies compared with the conventional body erase scheme.

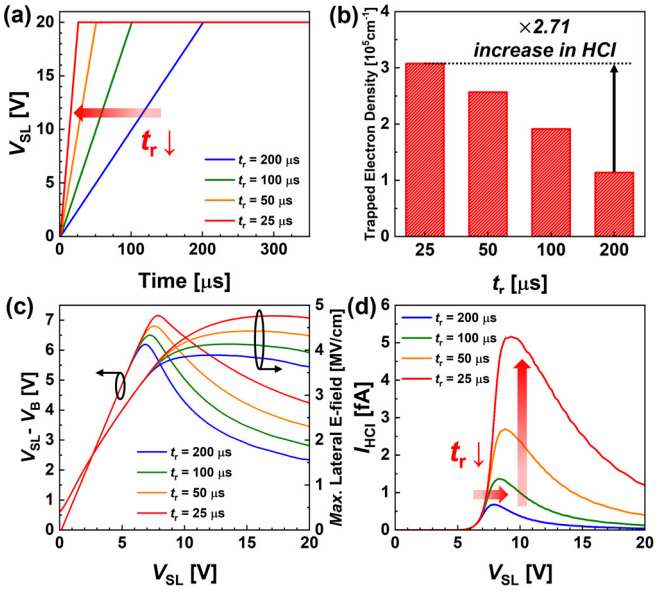


Fig. 7. (a) V_{SL} pulse of the GIDL erase operation with different rise times (t_r). (b) HCI-induced trapped electron density (Q_{trap}) in the SG CTL for different t_r values. Evolution of (c) $V_{SL} - V_B$ and E_m and (d) I_{HCI} during the rising phase for different t_r values.

A. SL Voltage Rise Time (t_r)

Fig. 7(a) shows the V_{SL} pulse with different rise times (t_r). As t_r decreases, resulting in a more rapid rise of the V_{SL} pulse, the HCI-induced trapped electron density (Q_{trap}) in the SG CTL increases, as shown in Fig. 7(b). This behavior arises because, for a shorter t_r, more holes must be generated per unit time and accumulated in the string to raise V_B in accordance with V_{SL}, requiring a higher GIDL-induced hole generation rate. As a result, larger $V_{SL} - V_B$ and E_m values are induced, with their peaks shifting to the right, as shown in Fig. 7(c). Consequently, I_{HCI} also increases, and the peak V_{SL} value shifts to a higher range, as shown in Fig. 7(d). A longer t_r helps suppress HCI; however, careful optimization is necessary due to the tradeoff with performance degradation. The use of a longer t_r slows down the erase operation speed of 3-D NAND, particularly in ISPE and subblock-level erase, where frequent erase operations are required. Moreover, as most I_{HCI} occur during the rising phase, a longer t_r corresponds to a prolonged stress time for generating Q_{trap}. It is also observed that the reduction of Q_{trap} is not linearly proportional to the increased t_r, as shown in Fig. 7(b). Even with an eightfold increase in t_r, Q_{trap} decreases by only ~2.71 times, indicating that the mitigation effect remains relatively limited.

B. Offset Length (L_{offset})

Fig. 8(a) shows the offset length (L_{offset}) between the SL junction and the SG gate edge. In 3-D NAND, offset structures have been employed to suppress the program disturbance by reducing the lateral electric field in the string [26]. Similarly, a longer L_{offset} is typically expected to reduce E_m and

Authorized licensed use limited to: Korea University. Downloaded on June 03,2026 at 07:47:29 UTC from IEEE Xplore. Restrictions apply.

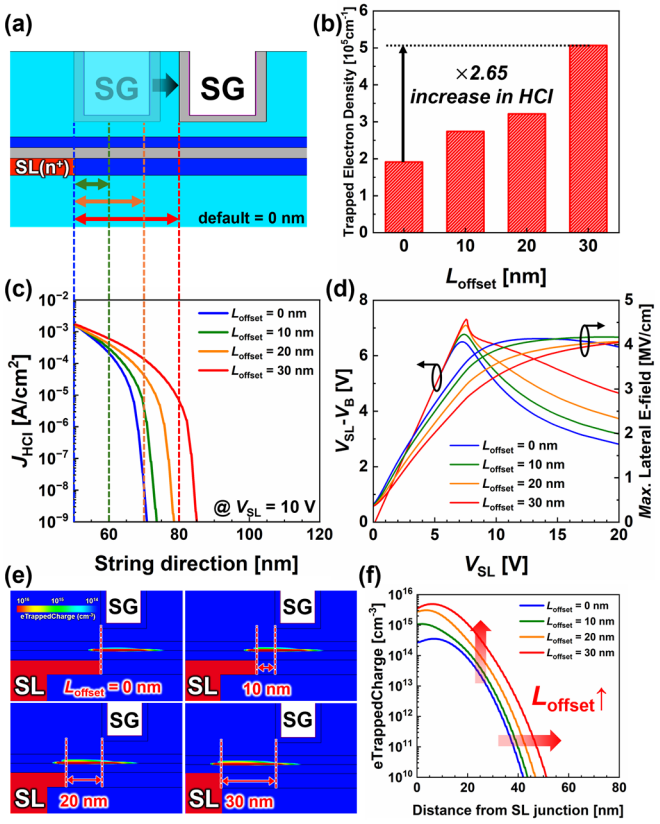


Fig. 8. (a) Offset length (L_{offset}) between the SL n^+ junction and the SG gate edge. (b) Q_{trap} in the SG CTL. (c) J_{HCI} in the string for different L_{offset} values. (d) Evolution of $V_{SL} - V_B$ and E_m during the rising phase for different L_{offset} values. (e) Spatial distribution of HCI-induced trapped electrons in the CTL. (f) Trapped electron density in the CTL along the distance from the SL junction for different L_{offset} values.

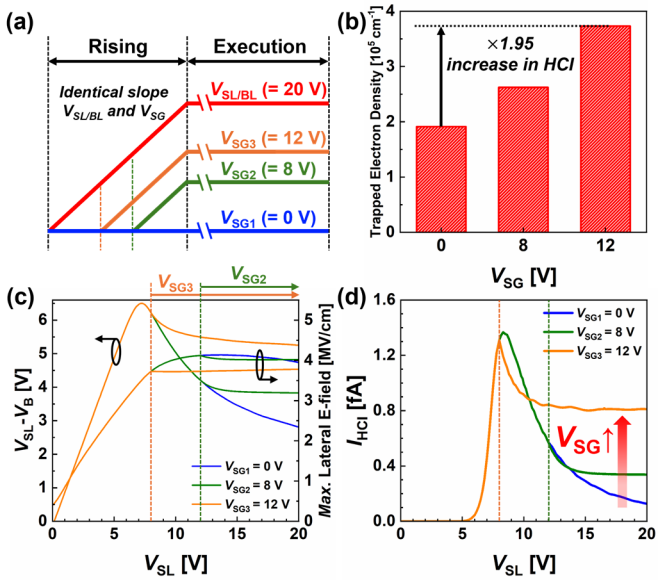


Fig. 9. (a) GIDL erase operation scheme, including V_{SG} bias. (b) Q_{trap} in the SG CTL for different V_{SG} values. Evolution of (c) $V_{SL} - V_B$ and E_m and (d) I_{HCI} during the rising phase for different V_{SG} values.

thereby mitigate the HCI degradation. However, it results in an increased Q_{trap} with a wider HCI-affected region, as shown in Fig. 8(b) and (c). This increased Q_{trap} originates from the direct involvement of the SG in the erase process. A reduction in E_m decreases the GIDL-induced hole generation rate in the SG, which in turn reduces hole accumulation in the string. Consequently, $V_{SL} - V_B$ increases and its peak shifts to the right, as shown in Fig. 8(d). The resulting higher $V_{SL} - V_B$, combined with weaker SG gate controllability under the offset region in the string, leads to a wider depletion width, which causes E_m to continue increasing during the rising phase. As a result, more HCI occurs across a wider region of the string with a longer L_{offset}. Fig. 8(e) and (f) shows the spatial distribution of HCI-induced trapped electrons in the CTL for different L_{offset} values. As L_{offset} increases, the trapped electron density not only becomes higher but also extends farther from the SL junction, thereby inducing a more significant shift in SG V_{th}. Therefore, this L_{offset} should be minimized to mitigate HCI while avoiding overlap between the SL junction and SG gate, which can also degrade erase efficiency [21]. These findings highlight that the precise control of the SL junction profile is critical in 3-D NAND fabrication.

C. Gate Bias of SG (V_{SG})

The impact of the gate bias of SG (V_{SG}) on HCI is analyzed, as shown in Fig. 9(a). During a practical erase operation, V_{SG} is applied to prevent unintended SG erase through FN tunneling by lowering the potential difference between the string and the SG gate ($V_B - V_{SG}$). In the conventional body erase scheme, a higher V_{SG} has been employed to reduce E_m and consequently mitigate HCI degradation [13]. In contrast, in the GIDL erase scheme, a higher V_{SG} aggravates HCI, as shown in Fig. 9(b), owing to the direct involvement of the SG in the erase process. When V_{SG} ramps with the same slope as V_{SL}, the V_{DG} of SG becomes constant. This fixed V_{DG} limits E_m, reducing the GIDL-induced hole generation rate and the resulting hole accumulation in the string. As a result, after V_{SG} is applied, $V_{SL} - V_B$ does not decrease rapidly and remains relatively high, as shown in Fig. 9(c). Furthermore, a higher V_{SG} lowers the effective barrier height toward the SG tunnel oxide for channel electrons, thereby facilitating the electron injection into the SG CTL [23]. Consequently, Fig. 9(d) shows that I_{HCI} also remains relatively high beyond its peak and increases further under a higher V_{SG} condition.

We also analyzed the effect of the SG gate bias rise time (t_{SG}), providing design guidelines for HCI mitigation. Fig. 10(a) shows the GIDL erase operation scheme, where V_{SG} ramps up with a shorter t_{SG} compared with the conventional case. A decrease in t_{SG} extends the V_{SL}-only ramping period ($t_r - t_{SG}$), thereby alleviating the constraint on E_m and enhancing the GIDL-induced hole generation rate. Consequently, as shown in Fig. 10(b), $V_{SL} - V_B$ after the rising phase becomes smaller with shorter t_{SG}, indicating mitigated HCI. This extended period also helps suppress the lowering of the effective barrier height for channel electrons caused by the applied V_{SG}. However, Fig. 10(b) also shows that when t_{SG} is shorter than 20 μs, a substantial trapped hole density in the SG CTL appears due to unintended FN tunneling-induced hole injection. To suppress such hole injection, the onset of the V_{SG} ramp should occur sufficiently early to reduce $V_B - V_{SG}$ before

Authorized licensed use limited to: Korea University. Downloaded on June 03,2026 at 07:47:29 UTC from IEEE Xplore. Restrictions apply.

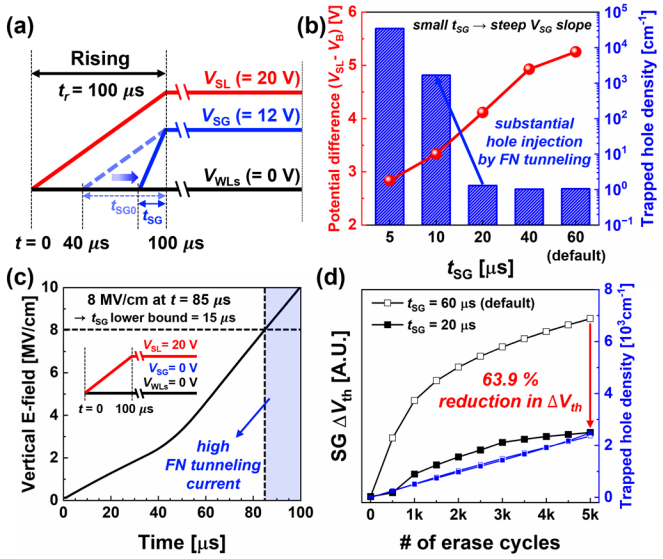


Fig. 10. (a) GIDL erase operation scheme with a shorter SG gate bias rise time (t_{SG}). (b) $V_{SL} - V_B$ after the rising phase and trapped hole density in the SG CTL for different t_{SG} values. (c) Evolution of the vertical electric field across the SG tunnel oxide under the V_{SG} = 0 V condition, used to monitor field evolution with increasing V_B. (d) HCI-induced SG V_{th} shift and trapped hole density in the SG CTL as a function of erase cycles.

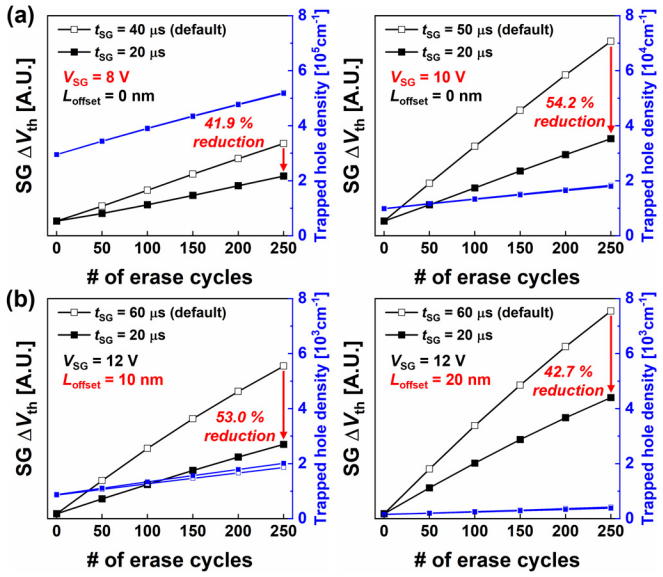


Fig. 11. HCI-induced SG V_{th} shift and trapped hole density in the SG CTL as a function of erase cycles under different (a) V_{SG} conditions, with the default t_{SG} corresponding to equal ramp slopes of V_{SG} and V_{SL}; and (b) L_{offset} conditions.

the vertical electric field across the SG tunnel oxide exceeds the critical level for FN tunneling (~8 MV/cm) [28], [29]. As shown in Fig. 10(c), the vertical electric field reaches 8 MV/cm at t = 85 μs during the rising phase under the V_{SG} = 0 V condition. Therefore, the V_{SG} ramp should begin before this time, resulting in a lower bound of $t_{SG} \approx 15$ μs.

Accordingly, the optimal t_{SG} is determined to be 20 μs with a slight design margin, which may be adjusted depending on the SG cell geometry and target V_{th} condition. Fig. 10(d) shows that the HCI-induced SG V_{th} shift is reduced by 63.9% after cycling stress under the t_{SG} = 20 μs condition, while the trapped hole density remains comparable to that of the default condition, confirming that FN tunneling-induced hole injection is also effectively suppressed. Furthermore, Fig. 11 demonstrates a consistent dependence of HCI and FN tunneling on t_{SG} under different V_{SG} and L_{offset} conditions, confirming the general validity of the proposed analysis.

V. Conclusion

This work presents a comprehensive investigation of HCI during the GIDL erase operation in 3-D NAND flash memory devices. The transient characteristics of HCI were analyzed in terms of two key physical parameters, $V_{SL/BL} - V_B$ and E_m, under the floating string condition. The dependence of HCI on critical design parameters revealed that a longer SL/BL voltage rise time, a shorter offset length, a lower SG gate bias, and a shorter SG gate bias rise time effectively mitigate HCI, demonstrating a strong correlation between HCI behavior and GIDL efficiency. The presented systematic analysis provides physical insights and design guidelines for HCI mitigation, thereby enhancing the reliability of state-of-the-art 3-D NAND flash memory technologies.

References

[1] S. I. Shim, J. Jang, and J. Song, "Trends and future challenges of 3D NAND flash memory," in *Proc. IEEE Int. Memory Workshop (IMW)*, May 2023, pp. 1–4, doi: 10.1109/IMW56887.2023.10145825.

[2] J. H. Kim et al., "Highly manufacturable 7th generation 3D NAND flash memory with COP structure and double stack process," in *Proc. Symp. VLSI Technol.*, Jun. 2021, pp. 1–2.

[3] H. Huh et al., "13.2 A 1Tb 4b/cell 96-stacked-WL 3D NAND flash memory with 30MB/s program throughput using peripheral circuit under memory cell array technique," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2020, pp. 220–221, doi: 10.1109/ISSCC19947.2020.9063117.

[4] J.-K. Kang et al., "Highly reliable cell characteristics with CSOB(channel-hole sidewall ONO butting) scheme for 7th generation 3D-NAND," in *IEDM Tech. Dig.*, San Francisco, CA, USA, Dec. 2021, p. 10, doi: 10.1109/IEDM19574.2021.9720643.

[5] H.-N. Yoo, Y. Yang, M.-K. Park, W.-Y. Choi, and J.-H. Lee, "Analysis of GIDL erase characteristics in vertical NAND flash memory," *J. Semicond. Technol. Sci.*, vol. 23, no. 3, pp. 196–201, Jun. 2023, doi: 10.5573/jsts.2023.23.3.196.

[6] C. Caillat et al., "3DNAND GIDL-assisted body biasing for erase enabling CMOS under array (CUA) architecture," in *Proc. IEEE Int. Memory Workshop (IMW)*, May 2017, pp. 1–4, doi: 10.1109/IMW.2017.7939067.

[7] H. Yoo et al., "Modeling and optimization of the chip level program disturbance of 3D NAND flash memory," in *Proc. 5th IEEE Int. Memory Workshop*, Monterey, CA, USA, May 2013, pp. 147–150, doi: 10.1109/IMW.2013.6582120.

[8] Y. Zhang et al., "Leakage characterization of top select transistor for program disturbance optimization in 3D NAND flash," *Solid-State Electron.*, vol. 141, pp. 18–22, Mar. 2018, doi: 10.1016/j.sse.2017.11.006.

[9] J.-D. Lee, C.-K. Lee, M.-W. Lee, H.-S. Kim, K.-C. Park, and W.-S. Lee, "A new programming disturbance phenomenon in NAND flash memory by source/drain hot-electrons generated by GIDL current," in *Proc. 21st IEEE Non-Volatile Semiconductor Memory Workshop*, Monterey, CA, USA, Feb. 2006, pp. 31–33, doi: 10.1109/.2006.1629481.

[10] M. Park et al., "NAND flash reliability degradation induced by HCI in boosted channel potential," in *Proc. IEEE Int. Rel. Phys. Symp.*, Anaheim, CA, USA, May 2010, pp. 975–976, doi: 10.1109/IRPS.2010.5488692.

[11] D. Son, J. Park, and H. Shin, "Investigation and compact modeling of hot-carrier injection for read disturbance in 3-D NAND flash memory," *IEEE Trans. Electron Devices*, vol. 67, no. 7, pp. 2778–2784, Jul. 2020, doi: 10.1109/TED.2020.2993772.

Authorized licensed use limited to: Korea University. Downloaded on June 03,2026 at 07:47:29 UTC from IEEE Xplore. Restrictions apply.

[12] H. Jo, J. Kim, and H. Shin, "A novel read scheme using GIDL current to suppress read disturbance in 3-D NAND flash memories," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 42, no. 12, pp. 5151–5155, Dec. 2023, doi: 10.1109/TCAD.2023.3281499.

[13] L. Yan et al., "Investigation of erase cycling induced TSG V_t shift in 3D NAND flash memory," *IEEE Electron Device Lett.*, vol. 40, no. 1, pp. 21–23, Jan. 2019, doi: 10.1109/LED.2018.2883770.

[14] J. H. Chang, J. H. Uhm, H. S. Kwon, E. Kwon, and W. Y. Choi, "Influence of channel hole remaining ratio on hemi-cylindrical vertical NAND flash memory," *IEEE Electron Device Lett.*, vol. 43, no. 9, pp. 1432–1435, Sep. 2022, doi: 10.1109/LED.2022.3192545.

[15] S.-H. Park et al., "Retention improvement in vertical NAND flash memory using 1-bit soft erase scheme and its effects on neural networks," in *IEDM Tech. Dig.*, Dec. 2022, pp. 5.5.1–5.5.4, doi: 10.1109/IEDM45625.2022.10019529.

[16] G. Ryu and M. Kang, "Selective erase operation for multiple strings of 3D ferroelectric (Fe)-NAND flash memory," *IEEE Electron Device Lett.*, vol. 45, no. 3, pp. 372–375, Mar. 2024, doi: 10.1109/LED.2023.3347519.

[17] S.-T. Lee et al., "Operation scheme of multi-layer neural networks using NAND flash memory as high-density synaptic devices," *IEEE J. Electron Devices Soc.*, vol. 7, pp. 1085–1093, 2019, doi: 10.1109/JEDS.2019.2947316.

[18] G. Malavena, A. L. Lacaita, A. S. Spinelli, and C. M. Compagnoni, "Investigation and compact modeling of the time dynamics of the GIDL-assisted increase of the string potential in 3-D NAND flash arrays," *IEEE Trans. Electron Devices*, vol. 65, no. 7, pp. 2804–2811, Jul. 2018, doi: 10.1109/TED.2018.2831902.

[19] *Sentaurus Device User Guide*, Synopsys, Mountain View, CA, USA, 2023.

[20] J. H. Chang, J. S. Woo, S.-K. Sung, K.-W. Song, and W. Y. Choi, "Ternary-state vertical NAND flash memory for the improvement of the density and accuracy of quantized neural networks," *IEEE Trans. Electron Devices*, vol. 71, no. 6, pp. 3985–3988, Jun. 2024, doi: 10.1109/TED.2024.3390651.

[21] J. Fan, M. Li, X. Xu, Y. Yang, H. Xuan, and R. Huang, "Insight into gate-induced drain leakage in silicon nanowire transistors," *IEEE Trans. Electron Devices*, vol. 62, no. 1, pp. 213–219, Jan. 2015, doi: 10.1109/TED.2014.2371916.

[22] G. A. M. Hurkx, H. C. de Graaff, W. J. Kloosterman, and M. P. G. Knuvers, "A new analytical diode model including tunneling and avalanche breakdown," *IEEE Trans. Electron Devices*, vol. 39, no. 9, pp. 2090–2098, Sep. 1992, doi: 10.1109/16.155882.

[23] K. Hasnat et al., "A pseudo-lucky electron model for simulation of electron gate current in submicron NMOSFET's," *IEEE Trans. Electron Devices*, vol. 43, no. 8, pp. 1264–1273, Aug. 1996, doi: 10.1109/16.506778.

[24] J. S. Woo, J. W. Lee, and W. Y. Choi, "Quantitative hot carrier injection analysis of N-type tunnel field-effect transistors," *IEEE Access*, vol. 11, pp. 14943–14950, 2023, doi: 10.1109/ACCESS.2023.3243578.

[25] D. Varghese et al., "Off-state degradation in drain-extended NMOS transistors: Interface damage and correlation to dielectric breakdown," *IEEE Trans. Electron Devices*, vol. 54, no. 10, pp. 2669–2678, Oct. 2007, doi: 10.1109/TED.2007.904587.

[26] W.-s. Cho et al., "Highly reliable vertical NAND technology with biconcave shaped storage layer and leakage controllable offset structure," in *Proc. Symp. VLSI Technol.*, Honolulu, HI, USA, Jun. 2010, pp. 173–174, doi: 10.1109/VLSIT.2010.5556216.

[27] G. Malavena, A. Mannara, A. L. Lacaita, A. S. Spinelli, and C. M. Compagnoni, "Compact modeling of GIDL-assisted erase in 3-D NAND flash strings," *J. Comput. Electron.*, vol. 18, no. 2, pp. 561–568, Jun. 2019, doi: 10.1007/s10825-019-01328-0.

[28] H. J. Kwo et al., "A valid experimental design of the lifetime prediction for NAND cell oxide," in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Grapevine, TX, USA, Apr. 2024, pp. 1–6, doi: 10.1109/IRPS48228.2024.10529448.

[29] Y. Taur and T. H. Ning, *Fundamentals of Modern VLSI Devices*, 3rd ed., Cambridge, U.K.: Cambridge Univ. Press, 2021.

Authorized licensed use limited to: Korea University. Downloaded on June 03,2026 at 07:47:29 UTC from IEEE Xplore. Restrictions apply.